Phan Nguyen Phan Anh

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Biography

My research focuses on multimodal AI and representation learning, integrating image analysis and vision-language models (VLMs) to solve problems across diverse application domains.

Education

- **University of Science and Technology of Hanoi (USTH)** 2022 – 2026
  Vietnam Academy of Science and Technology (VAST) Ha Noi, Vietnam
  - Degree: B.S. in Mechatronics Engineering Technology
  - GPA: 15.22/20 (French grading system) - Rank 1st among graduates of the major (Link)
  - Relevant coursework: Object-Oriented Programming (A), Introduction to Algorithms (A), C Programming (B+)
  - Thesis: Personalized Federated Learning via Similarity-Based Encoder Aggregation for Photovoltaic Power Forecasting

Experience

- **VinSmart Future** Dec 2026 (Incoming)
  AI Engineer (Intern)
  - Build and deploy multimodal LLM and RAG systems to production, with evaluation benchmarks to reduce hallucinations and optimize latency and cost.
- **Deep Learning and Application Group** Feb 2026 – Present
  Research Assistant
  - Working on Multimodal LLMs, NLP with RAG
  - Supporting the different research projects in the lab, contributing to research discussions and writing papers.
  - Advisor: Dr. Le Duc Anh
- **University of Science and Technology of Hanoi (USTH), VAST** Mar 2025 – Oct 2026
  Undergraduate Research Assistant Ha Noi, Vietnam
  - Conducted research on Federated Learning, Object Detection and Segmentation.
  - Developed novel framework for personalized federated learning in time-series forecasting domain.
  - Advisor: Dr. Vu Thi Tuyet Hong and Dr. Phan Anh Tuan

Publications

C=Conference, J=Journal, P=Preprint * indicates equal contribution.

Preprint/Under review

[P.1] N. P. A. Phan† (2026). Personalized Federated Learning via Similarity-Based Encoder Aggregation for Photovoltaic Power Forecasting. *Under review.*

Conference papers

[C.1] N. P. A. Phan, H. V. T. Tuyet†, V. H. Bui, A. T. Phan and V. C. Bui (2027). YOLO26-Guided SAM Prompting and Mask Selection for Lightweight PCB Defect Inspection. *The 12th International Conference on Intelligent Information Technology (ICIIT 2027).* [Accepted]

[C.2] N. P. A. Phan, T. X. T. Lam, and A. D. Le† (2026). Section-Aware RAG with Evidence Reranking for Biomedical Question Answering. *The 5th International Conference on Advances in Information and Communication Technology (ICTA).* [Accepted]

[C.3] H. V. T. Tuyet†, Q. D. Nguyen, A. T. Phan, **N. P. A. Phan**, A. D. Tran and V. C. Bui (2025). Global Horizontal Irradiance Prediction Using Interpretable Tiny Machine Learning. *The 2025 IEEE Asia Meeting on Environment and Electrical Engineering (EEE-AM).* [Paper]

Skills

- **Programming Languages:** Python, C, Java, LaTeX

- **Frameworks and Libraries:** PyTorch, Scikit-learn, NumPy, Pandas
- **Languages Proficiency:** Vietnamese (Native), English (TOEIC 815)

Honors and Awards

- University Scientific Research Contest (Third Prize)

Selected Certificates and Courses

- AI in Action 2026 (VinUniversity) – 12 weeks training onsite with $1,000 stipend.
- Deep Learning Specialization – Andrew Ng, Coursera
- Machine Learning – Andrew Ng, Coursera